

Euclidean Steinitz Bounds for Graver Bases and Integer Programming

Conditional structural bounds and algorithmic consequences

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Abstract

Let $A \in \mathbb{Z}^{m \times n}$ have maximum Euclidean column norm D . Assuming an absolute-constant Euclidean Steinitz bound $C\sqrt{m}$, we prove $g_1(A) \leq [K(1 + D)]^m$, where g_1 is the maximum ℓ_1 -norm of a Graver element and K is absolute. The proof counts lattice points in a Euclidean ball rather than its enclosing box. We obtain a bounded-variable LP–IP proximity bound of $(m + 1)[K(1 + D)]^m$, give explicit finite-state algorithms, and prove a two-level block-matrix bound $G[K(1 + D_A G)]^r$ with $G = [K(1 + D_B)]^s$, up to integer rounding. For columns with at most s nonzeros of magnitude at most Δ , the general bound specializes to $[O(1 + \sqrt{s}\Delta)]^m$. A chain of equations supplies a matching $D^{\Omega(m)}$ lower bound. We separately formulate a colorful consequence and explain why it does not by itself improve the dependence on the number of blocks in four-block integer programming. All main upper bounds are implications of an explicit geometric hypothesis, as claimed in a September 2026 preprint. We do not verify that preprint’s analytic proof or claim a new best general-IP runtime.

Draft status. This is an exploratory AI-generated manuscript. The arguments below are included in full where asserted as results. The Euclidean rearrangement input is isolated as an assumption; the colorful section has a separate input. Literature comparisons identify concrete earlier bounds, not an exhaustive priority determination. The four-block application remains a research direction, not a theorem of this paper. No computational solver speedup or formal verification is claimed.

1 Introduction

Steinitz rearrangements give a useful way to study integer kernels: expand a kernel vector into a zero-sum sequence of signed columns, reorder that sequence into a bounded region, and use a repeated partial sum to extract a smaller kernel vector. The geometry of the region matters as much as the rearrangement bound. In particular, a Euclidean ball of radius $O(\sqrt{m}D)$ contains at most $[O(1 + D)]^m$ integer points. Enclosing that ball in an axis-parallel box would introduce an avoidable factor of order $m^{m/2}$.

Our purpose is to develop this observation systematically for integer programming. Throughout, the number of rows is the dimension in which Steinitz is applied; it is not the number of variables. The principal matrix parameter is

$$D_2(A) := \max_{j \in [n]} \|A_{\cdot j}\|_2.$$

The familiar entry bound $\Delta = \max_{i,j} |A_{ij}|$ only implies $D_2(A) \leq \sqrt{m}\Delta$, and can conceal useful column geometry.

Results. Under Assumption 2.1, we prove a Graver bound $g_1(A) \leq Q_m(D_2(A))$, where $Q_m(D) = [O(1 + D)]^m$ is made explicit below. For an optimal vertex of a linear relaxation with integral variable bounds, we prove proximity at most $(m + 1)Q_m(D) - 1$. The proof handles fractional coordinates by separating the rearranged walk into runs with fixed lattice cosets. We then give a bounded-variable dynamic program, a separable-convex improvement oracle, and a standard-form state graph. For generalized N -fold matrices we apply the argument first to local kernel vectors and then to their global images. The resulting structural bound is independent of N and the brick widths. Finally, we give a balanced-subcollection consequence of a colorful Euclidean hypothesis.

Position relative to earlier work. Eisenbrand and Weismantel [2] established the Steinitz approach to proximity and integer-programming algorithms. Eisenbrand, Hunkenschröder, and Klein [3] used it to obtain the classical bound $g_1(A) \leq (2m\Delta + 1)^m$ and a two-level block bound. Our arguments retain their kernel-decomposition mechanism and change the geometry of the counting step. Precise comparisons appear in Section 8. Improvement over these particular bounds should not be confused with improvement over all later algorithms.

2 Geometric input and lattice counting

Assumption 2.1 (Euclidean rearrangement hypothesis). There is a fixed constant $C \geq 1$ such that every finite sequence $v_1, \dots, v_L \in \mathbb{R}^m$ with $\sum_i v_i = 0$ and $\|v_i\|_2 \leq D$ has a permutation π satisfying

$$\left\| \sum_{i=1}^k v_{\pi(i)} \right\|_2 \leq C\sqrt{m}D \quad (0 \leq k \leq L).$$

This holds for every $m \geq 1$ and $D \geq 0$, with repetitions permitted.

The release entitled *The Euclidean Steinitz–Bergström theorem* [1] claims this hypothesis and a stronger prescribed-order signing statement. We use the rearrangement assertion as a black box. The algorithmic statements below are uniform once a fixed integer upper bound on C is supplied. They do not require an algorithm for finding the asserted permutation.

Let B_2^m denote the closed Euclidean unit ball and set

$$\kappa = \sqrt{2\pi e}(C + 1), \quad Q_m(D) = \lceil [\kappa(1 + D)]^m \rceil. \quad (1)$$

Ceilings in this definition are only convenient explicit upper bounds; implementations may replace these expressions by larger integer bounds.

Lemma 2.2 (Uniform translated-ball count). *For every $a \in \mathbb{R}^m$ and $R \geq 0$,*

$$\#((a + \mathbb{Z}^m) \cap RB_2^m) \leq \frac{\pi^{m/2}}{\Gamma(m/2 + 1)} (R + \sqrt{m}/2)^m \leq \left[\sqrt{2\pi e} \left(\frac{R}{\sqrt{m}} + \frac{1}{2} \right) \right]^m. \quad (2)$$

In particular, the count is at most $Q_m(D)$ for $R = C\sqrt{m}D$.

Proof. Place a half-open unit cube centered at each of the counted lattice points. The cubes have disjoint interiors, each has volume one, and their union lies in $(R + \sqrt{m}/2)B_2^m$. This proves the first inequality. The estimate $\text{vol}(B_2^m) \leq (2\pi e/m)^{m/2}$ proves the second. Translation of the lattice has no effect on this argument. $\square$

Remark 2.3 (Rank and changes of coordinates). The parameter m can be reduced by deleting dependent rows; this preserves the integer kernel and cannot increase D_2 . For systems with right-hand sides one must first check consistency of the deleted equations. An arbitrary orthogonal change of coordinates does not preserve $\mathbb{Z}^m$. Thus applying Lemma 2.2 after such a rotation requires keeping track of the rotated lattice. We do not silently replace it by $\mathbb{Z}^m$.

3 Graver elements and a lower bound

For $u, v \in \mathbb{Z}^n$, write $u \sqsubseteq v$ if $u_i v_i \geq 0$ and $|u_i| \leq |v_i|$ for every i . The Graver basis $\mathcal{G}(A)$ consists of the $\sqsubseteq$ -minimal nonzero members of $\ker(A) \cap \mathbb{Z}^n$. Set $g_1(A) = \max\{\|g\|_1 : g \in \mathcal{G}(A)\}$, with value zero if the integer kernel is trivial. Every nonzero integer kernel vector is a sum of Graver elements conformal to it: repeatedly subtract a nonzero $\sqsubseteq$ -minimal kernel vector until the ℓ_1 -norm is zero.

Theorem 3.1 (Euclidean-column Graver bound). *Under Assumption 2.1, every $A \in \mathbb{Z}^{m \times n}$ satisfies*

$$g_1(A) \leq Q_m(D_2(A)). \quad (3)$$

Proof. Fix $g \in \mathcal{G}(A)$ and let $L = \|g\|_1$. Make a sequence containing $|g_j|$ copies of $\text{sign}(g_j)A_{\cdot j}$ for each nonzero coordinate. The sum is zero and each vector has norm at most $D = D_2(A)$. Use Assumption 2.1 and let p_k be the reordered partial sums. Then $p_0, \dots, p_{L-1}$ are integer points of $C\sqrt{m}DB_2^m$. They are distinct: if $p_i = p_j$ with $0 \leq i < j \leq L-1$, the intervening columns specify a nonzero integer vector $h \sqsubseteq g$ with $Ah = 0$ and $\|h\|_1 = j - i < L$. There is no coordinate cancellation because all copies of a given column have the sign prescribed by g . This contradicts Graver minimality. Lemma 2.2 now gives $L \leq Q_m(D)$. $\square$

Corollary 3.2 (Entry and sparsity parameters). *If $|A_{ij}| \leq \Delta$ and each column has at most $s \leq m$ nonzeros, then, under Assumption 2.1,*

$$g_1(A) \leq [O(1 + \sqrt{s}\Delta)]^m.$$

In particular the dense specialization is $[O(1 + \sqrt{m}\Delta)]^m$.

Proposition 3.3 (Exponential dependence cannot be removed). *For every integer $q \geq 2$ and $m \geq 1$ there is an integer matrix $A \in \mathbb{Z}^{m \times (m+1)}$ with at most two nonzeros per column such that*

$$D_2(A) \leq \sqrt{q^2 + 1}, \quad g_1(A) = 1 + q + \dots + q^m \geq q^m.$$

Proof. Use the equations $qx_i - x_{i+1} = 0$, $1 \leq i \leq m$. Their integer kernel consists exactly of integer multiples of $g = (1, q, \dots, q^m)$. Hence $\mathcal{G}(A) = \{g, -g\}$. The stated column bounds follow directly from the bidiagonal matrix. $\square$

For this family, writing $D_0 = \sqrt{q^2 + 1}$, the lower bound is at least $(2D_0/\sqrt{5})^m$. Thus the coefficient exponent m in Theorem 3.1 cannot be uniformly reduced for large D . This does not assert matching constants or extremality for every D .

3.1 An unconditional ℓ_1 comparison

It is important to compare against classical Steinitz in norms other than ℓ_∞ . Let $D_1 = \max_j \|A_{\cdot j}\|_1$. The arbitrary-norm Steinitz theorem (as stated in [3]) gives partial sums in $mD_1B_1^m$. Adding unit

cubes enlarges the ℓ_1 radius by at most $m/2$. Since $\text{vol}(B_1^m) = 2^m/m!$, the same proof gives the unconditional bound

$$g_1(A) \leq \frac{[2m(D_1 + 1/2)]^m}{m!} \leq [2e(D_1 + 1/2)]^m. \quad (4)$$

One can round the right-hand side upward if desired. In particular, $[O(1 + s\Delta)]^m$ already follows for s -sparse columns. The conditional Euclidean improvement replaces s by $\sqrt{s}$ in this comparison; single-exponential dependence for fixed s, Δ is not a new qualitative conclusion. Matrices such as incidence matrices can have much smaller Graver elements than either generic bound predicts.

4 Proximity with integral variable bounds

Consider

$$\max\{c^\top x : Ax = b, \ell \leq x \leq u, x \in \mathbb{R}^n\} \quad (\text{LP}),$$

$$\max\{c^\top z : Az = b, \ell \leq z \leq u, z \in \mathbb{Z}^n\} \quad (\text{IP}),$$

where A, b are integral and the finite entries of ℓ, u are integral. We assume an optimal LP vertex x^* and an optimal integer solution exist. The objective vector may be rational. Bounds may be infinite when these existence assumptions still hold.

Theorem 4.1 (Bounded-variable proximity). *Under Assumption 2.1, there is an optimal integer solution z^* such that*

$$\|z^* - x^*\|_1 \leq (m + 1)Q_m(D_2(A)) - 1. \quad (5)$$

More precisely, if x^* has f fractional coordinates, $m + 1$ may be replaced by $f + 1$.

Proof. At a vertex, at most $\text{rank}(A) \leq m$ coordinates are strictly between their bounds. Indeed, if there were more, their columns would admit a nonzero dependence supported on those coordinates, giving a sufficiently small feasible perturbation in both directions. A fractional coordinate cannot equal an integral finite bound, so $f \leq m$.

Choose an optimal integer solution z minimizing $\|z - x^*\|_1$. Such a minimizer exists since a bounded ball contains only finitely many integer points. Put $d = z - x^*$ and write

$$|d_j| = a_j + \theta_j, \quad a_j = \lfloor |d_j| \rfloor, \quad 0 \leq \theta_j < 1.$$

Form a sequence of a_j copies of $\text{sign}(d_j)A_{\cdot j}$, together with the vector $\theta_j \text{sign}(d_j)A_{\cdot j}$ when $\theta_j > 0$. The sum is zero. Call the latter vectors exceptional; there are exactly f of them. Every vector has Euclidean norm at most $D = D_2(A)$.

Reorder the sequence by Assumption 2.1. Exceptional vectors split the integral vectors into at most $f + 1$ consecutive runs, some possibly empty. During any one run all partial sums lie in a fixed coset of $\mathbb{Z}^m$ and in $C\sqrt{m}DB_2^m$. If a run contained $Q_m(D)$ or more integral vectors, it would have at least $Q_m(D) + 1$ partial sums, including its initial point. Lemma 2.2 would give two equal sums. The intervening integral vectors specify a nonzero integer h with $Ah = 0$, the same coordinate signs as d , and $|h_j| \leq a_j \leq |d_j|$.

Both $x^* + h$ and $z - h$ lie coordinatewise between x^* and z and are feasible for LP and IP, respectively. Optimality implies $c^\top h \leq 0$ from LP and $c^\top h \geq 0$ from IP. Thus $z - h$ is also integer optimal, whereas $\|z - h - x^*\|_1 = \|z - x^*\|_1 - \|h\|_1$, a contradiction.

Every run therefore has at most $Q_m(D) - 1$ integral vectors. Consequently

$$\|z - x^*\|_1 = \sum_j a_j + \sum_j \theta_j \leq (f + 1)(Q_m(D) - 1) + f = (f + 1)Q_m(D) - 1.$$

The weak inequality suffices, including the case $f = 0$. □

Remark 4.2 (Why fractional exceptions are kept). Replacing the fractional residual by an arbitrary short integral decomposition would require an additional argument and might increase the column norm. The proof above never does this. It counts each integral run in its own translated lattice, using the uniformity of Lemma 2.2.

The integrality gap is at most $\|c\|_\infty ((m+1)Q_m(D) - 1)$, by Hölder's inequality. For rational data this statement applies after the usual integral encoding of equality constraints; the resulting value of D must be computed for the actual encoded matrix.

Comparison with the Eisenbrand–Weismantel bound. For an entry bound Δ , their bounded-variable proximity theorem gives $m(2m\Delta + 1)^m$ [2]. Since $D_2(A) \leq \sqrt{m}\Delta$, Theorem 4.1 gives instead

$$(m+1)[O(1 + \sqrt{m}\Delta)]^m.$$

For $\Delta \geq 1$, the displayed dimension factor is reduced from m^m to $m^{m/2}$, up to absolute constants to the power m and the leading polynomial factor. The stronger column-norm formulation retains any additional saving from sparsity. This comparison is to that specific theorem; later determinant-based proximity results must also be considered, as discussed in Section 8.

5 Explicit algorithmic consequences

The bounds are useful without constructing a Steinitz order. A dynamic program may search a region guaranteed to contain an appropriate witness. Nevertheless, a structural bound is not automatically a running-time bound. We give explicit algorithms to make this distinction precise. All counts below concern arithmetic operations on numbers of polynomial encoding length, with the corresponding bit factors understood. Any fixed constant used to instantiate Assumption 2.1 is part of the algorithm specification, not something inferred from the input.

5.1 A bounded-variable exact dynamic program

Proposition 5.1. *Suppose all variable bounds are finite integers. Given an optimal LP vertex x^* , set $Q = Q_m(D)$ and $R = (m+1)Q + m$. Under Assumption 2.1, IP can be solved or declared infeasible using*

$$O\left(n(R+1)^2(2\lfloor DR \rfloor + 1)^m\right) \tag{6}$$

arithmetic operations in addition to solving LP. For nonzero integer A , the bit-time bound is

$$[O(1 + D)]^{O(m^2)} \text{poly}(n, L),$$

where L is the input length. LP infeasibility is handled first.

Proof. Let $w = \lfloor x^* \rfloor$ coordinatewise. Since the bounds are integral, w satisfies them, and $\|w - x^*\|_1 \leq m$. If IP is feasible, finite bounds imply existence of an optimum, and Theorem 4.1 gives an optimal solution $w + y$ with $\|y\|_1 \leq R$.

Process coordinates in order. Store the best value of $\sum_{j \leq i} c_j y_j$ for every state

$$(i, t, k), \quad t = \sum_{j \leq i} A_{.j} y_j \in \mathbb{Z}^m, \quad k = \sum_{j \leq i} |y_j| \leq R.$$

Each transition chooses an integer $y_i \in [-R, R]$ satisfying $\ell_i - w_i \leq y_i \leq u_i - w_i$. All reachable t satisfy $\|t\|_2 \leq DR$, hence each coordinate has absolute value at most $\lfloor DR \rfloor$. Initialize at $(0, 0, 0)$ and

accept states with $t = b - Aw$ and $k \leq R$. This gives the stated state and transition count. Every accepted solution is feasible, and an optimum is represented if one exists. If $A = 0$, independent coordinate optimization suffices. $\square$

This intentionally simple algorithm has a quadratic-in- m exponent. It must not be advertised as a $[O(1 + D)]^{O(m)}$ bounded-variable algorithm. Faster augmentation or convolution routines would require separate analysis.

5.2 Separable convex objectives: an improvement oracle

Let $F(x) = \sum_i f_i(x_i)$, where each f_i is discretely convex on its integer interval: $f_i(t+1) - f_i(t)$ is nondecreasing. Suppose evaluations and comparisons are available exactly.

Proposition 5.2. *Given a feasible integer point x with finite integer bounds, one can find an improving feasible point or certify optimality for F by searching only directions y with $Ay = 0$ and $\|y\|_1 \leq G$, where $G = Q_m(D)$ under Assumption 2.1. The search uses*

$$O\left(n(G+1)^2(2\lfloor DG \rfloor + 1)^m\right)$$

arithmetic operations and evaluations of objective differences.

Proof. Use the DP from Proposition 5.1, centered at x , with budget G , target zero, and transition cost $f_i(x_i + y_i) - f_i(x_i)$. Minimize this cost. If x is not optimal, let z be better and decompose $z - x$ into Graver elements $g^1, \dots, g^t$ conformal to $z - x$. Each $x + g^j$ satisfies the bounds. Discrete convexity, applied to monotone increments in each coordinate, gives

$$F(z) - F(x) \geq \sum_{j=1}^t (F(x + g^j) - F(x)).$$

Since the left-hand side is negative, at least one summand is negative. Theorem 3.1 ensures that direction is searched. $\square$

This is an exact optimality and improvement oracle, not a bound on the number of iterations of unit-step augmentation. Repeatedly taking an arbitrary improving step can have poor iteration complexity. In particular, we do not infer a new general MINLP algorithm: separability and discrete convexity are essential in this proposition.

5.3 Standard form and a Euclidean tube

Consider $Ax = b$, $x \in \mathbb{Z}_{\geq 0}^n$, and put $B = \|b\|_1$. Assume $D > 0$; for an integer matrix this implies $D \geq 1$.

Lemma 5.3. *Under Assumption 2.1, every feasible x admits an ordering of its multiset of columns whose partial sums lie in*

$$[0, b] + C\sqrt{m}DB_2^m. \tag{7}$$

Proof. Append M copies of $-b/M$ to the multiset, with $M = \max\{1, \lceil \|b\|_2/D \rceil\}$. All appended vectors have norm at most D , and the total is zero. Apply Assumption 2.1 and delete the appended vectors. A partial sum of the retained columns has the form $p + (k/M)b$, where $\|p\|_2 \leq C\sqrt{m}D$ and $0 \leq k \leq M$. $\square$

Proposition 5.4 (Explicit tube graph). *Under Assumption 2.1, standard-form feasibility has a finite graph algorithm with at most*

$$S \leq (B + 2)[K_0(1 + D)]^m \quad (8)$$

vertices and at most nS directed edges, for an absolute constant K_0 . After construction, breadth-first search takes $O(nS)$ operations. For a linear objective $c^\top x$, maximum-weight walks give either an optimum or an unboundedness certificate in $O(nS^2)$ operations. The graph can be constructed in $(B + 2)[K_1(1 + D)]^m \text{poly}(n, m, L)$ time for another absolute constant K_1 .

Proof. For $B > 0$, take the $B + 1$ centers $a_k = (k/B)b$, $0 \leq k \leq B$. For $B = 0$ take only the center zero. Adjacent centers have Euclidean distance at most one. Let the vertices be all integer points in the union of the balls of radius $T = C\sqrt{m}D + 1$ about these centers. This union contains the tube in (7). Lemma 2.2, with $T/\sqrt{m} + 1/2 \leq CD + 3/2$, proves (8) for a suitable K_0 . Include an edge $p \rightarrow p + A_j$, labeled j , whenever both endpoints occur. A walk from zero to b gives a feasible nonnegative integer solution by counting its labels. Lemma 5.3 proves the converse. Edges of zero displacement are allowed.

For completeness, the vertex enumeration need not visit an enclosing box. Enumerate coordinates of an integer point in a ball recursively, retaining a prefix only if its squared distance in the assigned coordinates is at most T^2 . Every retained prefix has an integer completion in the larger ball of radius $T + \sqrt{m}/2$: round the remaining center coordinates to nearest integers. At any fixed depth distinct prefixes have distinct such completions. Applying the cube argument again bounds the total number of prefixes over all depths by $(m + 1)[\sqrt{2\pi e}(CD + 2)]^m$. This proves the claimed construction bound, including deduplication and edge lookup up to polynomial factors. Radii may be enlarged to rational, polynomial-bit descriptions using a fixed upper bound on C , without changing these estimates.

Give an edge labeled j weight c_j . Every feasible solution yields a walk of the same objective value, and every walk yields a feasible solution. Consequently the graph represents exactly the attainable objective values. Restrict to vertices reachable from zero and able to reach b . A positive cycle there gives unbounded objective values; absent such a cycle, a maximum-weight walk can be taken simple after deleting nonpositive cycles. Bellman–Ford with negated weights gives the stated bound and detects the relevant positive cycles. $\square$

The dependence on B is numerical, not logarithmic. The proposition is an explicit geometric version of a state-graph argument, not a competitive replacement for algorithms that avoid this dependence. It also does not enforce upper bounds on label multiplicities, which is why bounded-variable IP was treated separately.

6 Two-level block matrices

Let $r, s \geq 1$ and consider

$$H = \begin{pmatrix} A_1 & A_2 & \cdots & A_N \\ B_1 & 0 & \cdots & 0 \\ 0 & B_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & B_N \end{pmatrix}, \quad A_i \in \mathbb{Z}^{r \times t_i}, \quad B_i \in \mathbb{Z}^{s \times t_i}.$$

The brick widths t_i need not be equal. Set

$$D_A = \max_{i,j} \|(A_i)_{\cdot j}\|_2, \quad D_B = \max_{i,j} \|(B_i)_{\cdot j}\|_2.$$

Theorem 6.1 (Two-level Euclidean bound). *Under Assumption 2.1, with $G = Q_s(D_B)$,*

$$g_1(H) \leq G Q_r(D_A G). \quad (9)$$

More generally, G can be any integer upper bound on $\max_i g_1(B_i)$ with $G \geq 1$, using Assumption 2.1 only in the r -dimensional global step.

Proof. Take $g \in \mathcal{G}(H)$ and partition it into bricks g_i . Since $B_i g_i = 0$, write each nonzero brick as a conformal sum $g_i = \sum_{j=1}^{q_i} h_{ij}$ with $h_{ij} \in \mathcal{G}(B_i)$. The bound on G gives $\|h_{ij}\|_1 \leq G$. The global images $v_{ij} = A_i h_{ij}$ satisfy

$$\sum_{i,j} v_{ij} = 0, \quad \|v_{ij}\|_2 \leq \sum_k |(h_{ij})_k| \|(A_i)_{\cdot k}\|_2 \leq D_A G.$$

There can be no nonempty proper zero-sum subcollection of these images. Such a subcollection would sum the corresponding local pieces to a nonzero integer kernel vector $h \sqsubseteq g$ different from g . The selected and omitted pieces are nonzero and conformal, so neither can cancel internally. This would contradict Graver minimality. The partial-sum argument of Theorem 3.1, now in dimension r , gives $\sum_i q_i \leq Q_r(D_A G)$. Hence $\|g\|_1 = \sum_{i,j} \|h_{ij}\|_1 \leq G Q_r(D_A G)$. $\square$

The formula does not depend on the number of bricks. It also permits sharper application-specific local bounds in place of $Q_s(D_B)$. For example, if a local block has a small explicitly known Graver basis, there is no reason to discard that information.

Comparison with the classical two-level calculation. For an entry bound $\Delta \geq 1$, the corresponding bound in [3, Lemma 3] is

$$G_{\text{old}}(2r\Delta G_{\text{old}} + 1)^r, \quad G_{\text{old}} = (2s\Delta + 1)^s. \quad (10)$$

Our dense substitution $D_B \leq \sqrt{s}\Delta$, $D_A \leq \sqrt{r}\Delta$ gives, with absolute constants displayed schematically,

$$G[K(1 + \sqrt{r}\Delta G)]^r, \quad G \leq [K(1 + \sqrt{s}\Delta)]^s. \quad (11)$$

Ignoring only factors $K^{O(rs+r+s)}$, the dimension factors in (10) are $s^{s(r+1)}r^r$ whereas those in (11) are $s^{s(r+1)/2}r^{r/2}$; the coefficient exponent in both comparisons is $s(r+1) + r$. This is a comparison of two structural bounds obtained by the same two-level decomposition, not an assertion about the best available complete N -fold algorithm.

If local columns have at most a nonzeros and global columns at most b , replace $\sqrt{s}$ and $\sqrt{r}$ by $\sqrt{a}$ and $\sqrt{b}$ in (11). However, the global images $A_i h_{ij}$ need not themselves be sparse. The proof uses their norm bound $D_A G$, not an unjustified assertion that they retain the original support size.

Algorithmic meaning. Theorem 6.1 bounds the total ℓ_1 -size of a Graver augmentation independently of N . It can be used as a structural input where an augmentation routine has an explicitly stated dependence on this size. Running the generic DP of Section 5 on all $r + Ns$ rows would lose the block advantage. A faster block-aware implementation and its full iteration bound are not supplied here.

7 Colorful sequences and what they do not yet prove

The ordinary zero-sum hypothesis is sufficient for all preceding results. This section uses the following distinct input, also claimed as a consequence in the released preprint [1].

Assumption 7.1 (Colorful Euclidean hypothesis). For k sequences $v_{j,1}, \dots, v_{j,L} \in \mathbb{R}^d$, $1 \leq j \leq k$, with $\|v_{j,i}\|_2 \leq D$ and $\sum_{j,i} v_{j,i} = 0$, there are permutations π_j such that

$$\left\| \sum_{j=1}^k \sum_{i=1}^t v_{j,\pi_j(i)} \right\|_2 \leq C_{\text{col}} \sqrt{d} D \quad (0 \leq t \leq L),$$

where C_{col} is absolute and independent of k, L .

The source obtains $C_{\text{col}} = 2C$ from its stronger signing result and matrix transference. We do not claim that Assumption 7.1 has been proved here from Assumption 2.1 alone. Define

$$Q_d^{\text{col}}(D) = \left\lceil [\sqrt{2\pi e}(C_{\text{col}} + 1)(1 + D)]^d \right\rceil.$$

Definition 7.2. An array of integer vectors $v_{j,i}$ with k rows and L entries per row, whose total sum is zero, is *balanced-indecomposable* if there are no subsets $I_j \subseteq [L]$ of a common cardinality $t \in \{1, \dots, L-1\}$ such that $\sum_j \sum_{i \in I_j} v_{j,i} = 0$.

Proposition 7.3. Under Assumption 7.1, a balanced-indecomposable array with $\|v_{j,i}\|_2 \leq D$ has

$$L \leq Q_d^{\text{col}}(D). \quad (12)$$

Proof. Let p_t be its synchronous partial sums after the row permutations. The L integer points $p_0, \dots, p_{L-1}$ lie in $C_{\text{col}} \sqrt{d} D B_2^d$. A repetition $p_a = p_b$, $a < b < L$, selects $b - a$ entries from each row with total zero, contrary to the definition. The translated-ball count proves the assertion. $\square$

The bound controls the number of rounds L , not the total number kL of vectors. This distinction is essential. With $L = 1$, any number of rows can have total zero, and the balanced-indecomposability condition has no proper positive number of rounds to exclude. Thus the proposition cannot alone remove the number of colors from a Graver norm bound.

Proposition 7.4 (Centering when the total is nonzero). Under Assumption 7.1, an arbitrary array of the same shape and vector bound admits row permutations for which, writing $T = \sum_{j,i} v_{j,i}$,

$$\left\| \sum_{j=1}^k \sum_{i=1}^t v_{j,\pi_j(i)} - \frac{t}{L} T \right\|_2 \leq 2C_{\text{col}} \sqrt{d} D \quad (0 \leq t \leq L).$$

Proof. Subtract the mean $\mu_j = L^{-1} \sum_i v_{j,i}$ from every entry in row j . Each centered vector has norm at most $2D$ and the total is zero. Apply Assumption 7.1 and then restore the means. $\square$

Four-block IP remains an application to complete. Oertel, Paat, and Weismantel [6] develop colorful rearrangements for block-structured proximity. Bárány [7] later improves the arbitrary-norm row-permutation bound to $4d - 2$. Therefore a comparison only against the earlier $40d^5$ bound would overstate what is new. With entrywise magnitude M , Euclidean conversion gives $D \leq \sqrt{d}M$, so the new rearrangement radius is $O(dM)$, the same coordinate scale as the later arbitrary-norm

bound. An additional saving can come from ball counting, provided the proof really supplies integer partial sums or a controlled number of lattice cosets.

To obtain a full four-block theorem, one must verify three further facts inside that decomposition: the Euclidean norms of the derived vectors, the precise balanced selection that yields an admissible kernel move, and the number of exceptional shifts or rounds that need separate counting. Neither Proposition 7.3 nor Proposition 7.4 establishes these facts. We therefore make no claim here of an improved polynomial exponent in the number of blocks.

8 Comparisons and limits of the conclusions

Dense general-IP algorithms. Jansen and Rohwedder [4] already give algorithms with the $\sqrt{m}\Delta$ base, and formulate their analysis using hereditary discrepancy. Thus the dense specialization of our bound does not by itself improve the known general-IP runtime. Our tube algorithm also retains a numerical dependence on $\|b\|_1$. Its role is to show constructively how the geometric hypothesis can certify a smaller state space, rather than to establish a best running time.

Determinant-based proximity. For full row rank, let $\Delta_m(A)$ be the largest absolute determinant of an $m \times m$ submatrix. In nonnegative standard form, Lee, Paat, Stallknecht, and Xu [5] prove proximity less than

$$3m^2 \log(2\sqrt{m} \Delta_m(A)^{1/m}) \Delta_m(A).$$

Hadamard gives $\Delta_m(A) \leq D_2(A)^m$. Their result can therefore be stronger than our direct bound on the overlapping formulation. Theorem 4.1 includes integral upper and lower bounds, but we do not infer novelty merely from this difference without checking the applicable extensions. It is a complete implication of the geometric hypothesis and a convenient direct proof, not a universal improvement.

What the sparse comparison establishes. For fixed s, Δ , both (4) and Corollary 3.2 give $2^{O(m)}$. For growing s , the displayed Euclidean upper bound replaces a base proportional to $s\Delta$ by one proportional to $\sqrt{s}\Delta$. The constants are absolute and inherited from the geometric hypothesis. This comparison does not establish that every matrix with those parameters has Graver elements near either upper bound.

Practical interpretation. Smaller structural bounds can reduce a certified search region, but hidden constants, state representation, preprocessing, and instance structure all matter. No implementation or benchmark in this manuscript supports a claim of faster commercial branch-and-bound or cutting-plane solvers. Likewise, the convex improvement oracle does not extend to arbitrary nonconvex objectives. A useful experimental follow-up would compare actual reachable states in Euclidean and coordinate-box neighborhoods for selected bounded-column instances.

9 Further questions

The most immediate theoretical task is a comprehensive novelty audit of the Euclidean-column Graver bound, including bounds expressed through determinants, circuits, discrepancy, and other column norms. The lower example proves the necessity of exponential growth in m and the coefficient scale, but does not settle the best absolute base.

A second task is a full block-aware augmentation analysis using Theorem 6.1. The structural bound should be substituted into an explicitly chosen algorithm, including step lengths, feasibility

initialization, and the number of augmentation phases. Doing so would determine whether the dimension-factor improvement survives all other terms in the runtime.

A third task is the four-block program described in Section 7. Its main difficulty is preserving the correct integer or translated-lattice structure through the colorful decomposition. An improvement in a rearrangement radius alone does not settle it.

Finally, one can ask for anisotropic versions: choose a norm adapted to the columns and count the actual image lattice in the resulting body. Any such result must control both the body and the lattice after preconditioning. Optimizing only the column norm while forgetting the lattice density would give an invalid improvement.

A Dependency and verification ledger

The logical dependencies of the stated results are as follows.

Result	External geometric input
Translated-ball count; chain lower bound	None
Unconditional ℓ_1 comparison Graver bound; proximity; state algorithms	Classical arbitrary-norm Steinitz theorem Assumption 2.1
Two-level block bound	Assumption 2.1 locally and globally, or a supplied local bound
Balanced-array bound; centering	Assumption 7.1
Improved four-block IP theorem	Not proved in this draft

The proofs were developed as implications of the stated assumptions. The analytic proof in the September 2026 release has not been independently verified here. The release is a mutable online source; the bibliography records its dated directory and the access date. Before submission, an archived revision or commit should be fixed, author metadata verified, and any available corrections checked. No Lean verification is included. Finite examples or numerical experiments, if added, cannot replace the universal arguments above. The repository accompanying this draft records compilation and any small-instance checks separately.

References

- [1] OpenAI math preprint release, *The Euclidean Steinitz–Bergström theorem*, dated directory September 24, 2026. [Repository source](#). The statements used here are in `build/introduction.tex` and `build/consequences.tex`. Accessed October 7, 2026. Repository attribution is used here without assigning individual authorship.
- [2] F. Eisenbrand and R. Weismantel, *Proximity results and faster algorithms for Integer Programming using the Steinitz Lemma*, [arXiv:1707.00481](#), 2017; version 3, 2019.
- [3] F. Eisenbrand, C. Hunkenschroder, and K.-M. Klein, *Faster Algorithms for Integer Programs with Block Structure*, [arXiv:1802.06289](#), 2018.
- [4] K. Jansen and L. Rohwedder, *On Integer Programming, Discrepancy, and Convolution*, [arXiv:1803.04744v4](#), 2022.

- [5] J. Lee, J. Paat, I. Stallknecht, and L. Xu, *Improving proximity bounds using sparsity*, [arXiv:2001.04659](#), 2020.
- [6] T. Oertel, J. Paat, and R. Weismantel, *A Colorful Steinitz Lemma with Application to Block-Structured Integer Programs*, [arXiv:2201.05874v2](#), 2022.
- [7] I. Bárány, *A matrix version of the Steinitz lemma*, [arXiv:2308.10102v2](#), 2024.